Project 6: Step-by-Step Guide

What each member does, day by day, with who to send each result to

How to use this guide

Find your member number and follow your steps in order. The **Send to** column says who receives your output. Shaded rows are **team checkpoints**: if you can't meet one, tell the group that day. **Handover rule:** code goes to **M1**; experiment results go to **M2**; report text and slides go to **M6**.

Team checkpoints

Date	What must be true	Owner	Done when
18 Oct	Kickoff call: roles, settings and plan confirmed.	All	Everyone has MADE installed.
20 Oct	Agent skeleton runs one full episode; baseline numbers exist.	M1, M2	Demo in group chat.
24 Oct	**Handover 1:** code to M1; method text and draft slides to M6.	M2–M5	M1 and M6 confirm receipt.
25 Oct	**Freeze the agent.**	M1	Version tagged in GitHub.
26 Oct	Reference run shared; all ablations and hidden-test runs started.	M1, all	Reference sent to M2 by noon.
27 Oct	**Handover 2:** all results to M2; results text and slides to M6.	All	M2 and M6 confirm receipt.
29 Oct	Full rehearsal; report and slides final.	All	Timed under 10 min.

Everyone: first steps (17–18 Oct)

Date	Step	Send to	Done when
17 Oct	Read the project brief and skim the MADE paper's introduction (arxiv.org/abs/2601.20996).	—	You can explain the agent → oracle → repeat loop.
17 Oct	Pick roles: M1 to the strongest coder, M2 to the second strongest.	Group chat	Every role has a name.
17 Oct	Create a free Materials Project account and get an API key.	—	Key saved.
18 Oct	Clone github.com/diffractivelabs/MADE, run `uv sync`, add keys to `.env`, run `uv run scripts/run_benchmark.py`. Time how long it takes.	Group chat (run time)	Results folder appears.
18 Oct	Kickoff call: confirm with the TA the repo version, base AI model, query budget, repeat runs, and what is due on 30 Oct. Agree who pays for LLM API and cloud costs.	Group chat	Settings written down.
18 Oct	Start your work log.	—	First entry written.

Member 1: System integration

Builds the agent everyone plugs into, merges all code, and designs how the budget is spent.

Date	Step	Send to	Done when
19 Oct	Read `src/made/agents/README.md` and the Hydra configs in `configs/`. Create the shared GitHub repo, one branch per member.	Group chat	Everyone has repo access.
20 Oct	Build the agent skeleton with slots for planning (M3), memory (M4), exploration (M5), tool routing (M2), each with an on/off setting. Run one full episode.	Group chat (demo)	**Checkpoint:** one episode runs end to end.
21 Oct	Write a one-page 'how to plug in your part' note.	M2–M5	They confirm they understand it.
21–23 Oct	Design and build budget management: how many queries go to each round, and when to stop a weak direction.	—	Budget logic runs in the agent.
24 Oct	Receive and merge everyone's code. Make every run save trajectory, tokens and run time. Send your method text and draft slides.	M6 (text, slides)	All parts run together.
25 Oct	**Freeze the agent** and tag the version.	Group chat	Tag exists.
26 Oct	Run the full frozen agent once on the practice materials (ablation budget and agreed seeds). This is everyone's reference.	M2 (by noon)	Reference numbers sent.
26–27 Oct	Write the reproduction README: commands, configs, seeds, dependency versions. Help M6 with compute for hidden-test runs.	M6 (text, slides by 27 Oct)	Someone else can rerun one result.
28–29 Oct	Review your report section; rehearse: architecture and budget management.	—	Timed under 2 min.

Member 2: Baselines and evaluation

Produces the scoreboard everyone compares against, and designs how the agent chooses its tools.

Date	Step	Send to	Done when
19 Oct	Run the Random and Chemeleon baselines with `scripts/run_baseline_experiments.py` on the practice materials.	—	Baseline results saved.
20 Oct	Build one script (using MADE's `notebooks/` utilities) that outputs mSUN@B, AUDC, AF, EF and discovery curves. Help M1 if setup is stuck.	Group chat	**Checkpoint:** baseline table and curve shared.
21–23 Oct	Design and build tool routing: rules for when the agent uses which generator, filter or scorer. Test on practice materials.	—	Tool routing runs in your slot.
24 Oct	Send your code and your method text and draft slides.	M1 (code), M6 (text)	Both confirm receipt.
26 Oct	Receive M1's reference. Run the tool routing ablation (routing off vs. reference).	—	Ablation running.
27 Oct	Collect everyone's results. Build the final results table (us vs. both baselines), discovery curves and cost analysis.	M6 (by end of day)	Table, curves and your results text sent.
28–29 Oct	Review your report section; rehearse: results, curves, tool routing.	—	Timed under 2 min.

Member 3: Planning and self-correction

Designs how the agent plans its research and recovers when it gets stuck.

Date	Step	Send to	Done when
19–20 Oct	Design planning (plan → propose batch → review → update plan) and stuck detection (e.g. no new valid material in the last K queries → re-plan). Agree the slot interface with M1.	M1	One-page design agreed.
21–23 Oct	Build both in your slot; test on 1–2 practice materials.	—	Both run in the agent.
24 Oct	Send your code and your method text and draft slides.	M1 (code), M6 (text)	Both confirm receipt.
26–27 Oct	Two ablations vs. M1's reference: planning off; self-correction off. Own laptop, ablation budget.	M2 (results)	Results sent by 27 Oct.
27 Oct	Write your results text (what changed and why) and finish your slides.	M6	Sent by end of day.
28–29 Oct	Review your report section; rehearse.	—	Timed under 2 min.

Member 4: Memory

Designs what the agent remembers, and keeps the evidence of what it discovered.

Date	Step	Send to	Done when
19–20 Oct	Design memory: what to store per tested material (composition, oracle result, why it failed) and how to summarize it to fit the AI model's prompt.	M1	One-page design agreed.
21–23 Oct	Build memory and the evidence log. Agree the log format with M5.	M5 (log format)	Memory runs; M5 confirms format.
24 Oct	Send your code and your method text and draft slides.	M1 (code), M6 (text)	Both confirm receipt.
26–27 Oct	Memory ablation vs. M1's reference. Pick 2–3 example materials the agent discovered.	M2 (results)	Results and examples sent by 27 Oct.
27 Oct	Write your results text and finish your slides.	M6	Sent by end of day.
28–29 Oct	Review your report section; rehearse.	—	Timed under 2 min.

Member 5: Exploration

Designs how the agent balances new areas against promising ones, and documents how it fails.

Date	Step	Send to	Done when
19–20 Oct	Design the exploration policy: share of budget for new areas vs. refining near successes, and whether it changes as the budget runs out.	M1	One-page design agreed.
21–23 Oct	Build the policy; test on practice materials; note runs that look like over-exploitation or wasteful exploration.	—	Policy runs; candidate examples noted.
24 Oct	Send your code and your method text and draft slides.	M1 (code), M6 (text)	Both confirm receipt.
26–27 Oct	Exploration ablation vs. M1's reference. Using M4's logs, pick and chart one trajectory per failure mode.	M2 (results)	Results and 2 failure examples by 27 Oct.
27 Oct	Write the failure section and finish your slides.	M6	Sent by end of day.
28–29 Oct	Review your report section; rehearse.	—	Timed under 2 min.

Member 6: Experiment design, generalization and report lead

Designs how we test fairly, runs the hidden-test comparison, and turns everyone's work into one report and deck.

Date	Step	Send to	Done when
17–18 Oct	Fill in the Group C spreadsheet. Ask the TA: hidden-test release date, and what exactly is due on 30 Oct.	Course / TA	Sheet filled; TA asked.
18 Oct	Create the report and slide templates with every required section, plus a requirements checklist.	Group chat	Templates shared.
19–20 Oct	Design the validation protocol: split practice materials into tuning and check sets; fix seeds, budgets and repeat runs.	Group chat	Protocol shared.
21–23 Oct	Run check-set tests as parts come in, to catch overfitting early.	—	Check-set results logged.
24 Oct	Receive everyone's method text and draft slides; slot them into the templates.	—	All 5 received.
26–27 Oct	Run the frozen agent and both baselines on the hidden test materials (M1 helps with compute). If not released, use the check set and say so.	M2 (results by 27 Oct)	Per-material table done.
28 Oct	Combine all ablations (which part helped most, and why); write the discussion; assemble the full report and slides.	Group (for review)	Full draft shared.
29 Oct	Run the rehearsal; apply final edits.	—	Report and slides final.
30 Oct	Submit what's due (as confirmed with TA).	Course	Submitted.

Everyone: after submission

Date	Step	Send to	Done when
31 Oct / 1 Nov	Present online: everyone presents their own part and answers Q&A.	—	Done.
1–3 Nov	Write your individual report (max 3 pages) from your work log: Contributions, Lessons Learned, Future Plans (optional extensions like multi-agent fit here). Explain *why* your part changed the score.	YAOLONG001@e.ntu.edu.sg	Sent by end of 3 Nov.